

COURSE DESCRIPTION			
Program	Artificial Intelligence, Artificial Intelligence & Machine Learning, Cloud Computing and Big Data, Cyber Forensics and Information Security, Computer Science and Technology, Computer Science & Engineering, Computer Engineering, Computer Science and Engineering (Artificial Intelligence & Machine Learning), Information Technology, Automation and Robotics, Robotic Process Automation		
Course Title	Mathematics for Computer Technology		
Course Code	2001C	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	3:1:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course introduces the essential mathematical concepts for engineering studies. It covers the fundamentals of complex numbers, integral calculus, Boolean Algebra, Logic, Statistics and Graph Theory. These topics are meant to give the basic knowledge to answer a wide range of engineering problems in computer engineering and related fields. Emphasis is placed on developing skills for solving engineering problems.

Course Objectives

1	To provide comprehensive introductory-level coverage of complex numbers.
2	To introduce the fundamentals of integral calculus to solve engineering-related problems.
3	To provide the basic concepts of logical and statistical operations for data analysis and processing.
4	To provide an introduction to graph theory.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Trigonometry, Differentiation, Matrices	1002	Fundamentals of Engineering Mathematics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply concepts of complex numbers to solve problems.	Apply
CO2	Use integration techniques to solve basic engineering problems.	Apply
CO3	Perform data analysis using the foundation of logic and statistical tools.	Apply
CO4	Apply basic graph theory concepts to represent and solve real-world problems.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Total	12	-	-	-	-	-	-	-	-	-	-
Correlation Level	3	-	-	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
							CIE	ESE	Total	CIE	ESE	Total
L	T	P	S									

3	1	0	0	5.5	60	4	40	60	100	0	0	0	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Complex Numbers	The Complex Number System, Fundamental operations with complex numbers, Modulus and Amplitude, Graphical representation of complex numbers, Polar form	11	4
Module II	Integral Calculus	Integration as the reverse process of differentiation, Standard results, Integration by parts, Integration by substitution of $f(ax+b)$, $f(x) * f'(x)$ and $f'(x)/f(x)$, Definite integrals, Area bounded by a curve and the x-axis.	11	4
Module III	Logic and Basic Statistics	Logic: Propositions, Logical operations: Negation, Conjunction, Disjunction and Exclusive OR of propositions and their truth tables. Tautology, contradiction and contingency. Statistics: Introduction to Statistics, Grouped data, Frequency distribution, Measures of Central Tendency.	11	4
Module IV	Graph Theory	Basic Concepts and Examples, Paths, Walks, Cycles and Connectivity, Trees and Spanning Trees	12	3

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
The complex number system	Understand the complex number system and identify the real part and imaginary part	L	CO1	Understand	1
Conjugate of a complex number, Equality of complex numbers	Determine the conjugate of a complex number, Test the equality of complex numbers by comparing their real and imaginary parts.	L	CO1	Apply	1
Fundamental operations with complex numbers : Addition, Subtraction, Multiplication, Division	Perform fundamental operations on complex numbers.	L	CO1	Understand	5
Graphical Representation	Represent complex numbers graphically on the Argand (complex) plane.	L	CO1	Understand	1
Absolute value and Amplitude of a Complex Number	Determine the Modulus (Absolute value) and amplitude (argument) of a complex number in degrees.	L	CO1	Apply	2
Polar Form	Express a complex number in polar form	L	CO1	Understand	1
Problems	Solve simple engineering problems involving complex numbers.	T	CO1	Apply	4
Module II					
Integration as the reverse process of differentiation	Explain integration as the inverse process of differentiation	L	CO2	Remember	1
Integral of standard functions(x^n , e^x , $1/x$, $\sin x$, $\cos x$, $\sec^2 x$, $\operatorname{cosec}^2 x$, $\sec x \tan x$, $\operatorname{cosec} x \cot x$)	Define standard integrals of basic functions	L	CO2	Remember	1
Rules of integration and simple problems, Integration by substitution $f(ax+b)$, $f(x) * f'(x)$ and $f'(x)/f(x)$	Use basic rules of integration (constant multiple, sum and difference rules) to solve simple integrals.	L	CO2	Understand	2
Integration by parts rule and problems	Apply the integration by parts method to evaluate integrals of product of two functions	L	CO2	Apply	2
Definite integrals and simple problems	Evaluate definite integrals	L	CO2	Apply	2
Application of definite integrals	Determine area bounded by a curve and the x-axis	L	CO2	Apply	3
Problems	Evaluate integrals	T	CO2	Apply	4

Module III					
Propositions: Definition and notation, examples, truth value of propositions, compound propositions	Form propositions and distinguish propositions and non propositions	L	CO3	Understand	1
Logical operations: Negation, conjunction, disjunction and exclusive or of propositions and truth tables	Apply Negation, Conjunction, Disjunction, and Exclusive OR and construct the respective truth tables.	L	CO3	Apply	3
Tautology, contradiction and contingency	Analyze logical expressions and identify tautology contradiction and contingency.	L	CO3	Apply	3
Introduction to Statistics, grouped data	Define statistics, organize grouped data, and construct frequency distributions.	L	CO3	Understand	2
Finding the mean, median and mode from a grouped frequency distribution	Calculate mean, median, and mode from an ungrouped frequency distribution.	L	CO3	Apply	2
Problems	Solve problems related to Logic and statistics	T	CO3	Apply	4
Module IV					
Graph, Vertex, Edge, Degree of a Vertex, Vertex Set, Edge Set, Loops, Simple Graph.	Explain the basic concepts of graphs and use them to model simple real-life situations.	L	CO4	Understand	2
Statement of Handshaking Lemma and its corollary (Without proof), Subgraphs, Union of graphs.	Explain the union of graphs, subgraphs, and the basic relationship between vertices and edges in a graph.	L	CO4	Understand	2
Matrix representation of graph, Adjacency matrix, Incidence matrix.	Apply the matrix representation of graphs and construct graphs from adjacency and incidence matrices.	L	CO4	Apply	2
Null Graph, Complete Graph, Path Graph, Cycle Graph, Connected graph, Regular Graph, Weighted Graph, Tree, Spanning Tree.	Identify and classify common types of graphs such as null, complete, cycle, path, regular, weighted graphs and trees.	L	CO4	Understand	3
Dijkstra's algorithm, Kruskal's Algorithm	Apply Dijkstra's algorithm to solve shortest path problems and Kruskal's algorithm to find minimum spanning trees.	L	CO4	Apply	3
Problems	Apply graph theory concepts and algorithms to solve related problems.	T	CO4	Apply	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Find the conjugate of complex numbers, determine modulus, amplitude and write the polar form of complex numbers	4
2	Solve problems related to equality, addition and subtraction of complex numbers	4
3	Multiply and divide complex numbers	4
4	Find the applications of complex numbers in respective engineering areas.	4
Module II		
5	Evaluate integrals of functions using basic rules	4
6	Evaluate integrals by the method of substitution	4
7	Solve problems by applying integration by parts.	4
8	Apply the definite integral technique to find the area of a region	3
Module III		
9	Construct truth table related to Real-Life problems	4
10	Solve the problems of identifying whether each expression is a tautology, contradiction or contingency	4
11	Problems of finding mean, median and mode from a grouped data.	4
12	Find the applications of statistics in respective engineering areas.	3
Module IV		
13	Draw real-life examples of graphs (road maps, networks, computer links); label vertices/edges and count degrees.	4

14	Sketch simple paths/cycles in given small graphs. Check connectivity by tracing paths.	4
15	Find disconnected graphs in daily life (e.g., separate computer networks) and draw them; identify other types of graphs (bipartite, complete bipartite graphs etc.)	4
16	Find the applications of graph theory in respective engineering areas.	3
Total Self-Learning hours		60

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Mathematics for Computer Technology		SITTTR Kalamassery

REFERENCE			
Sl. No.	Title	Author	Publication
1	Schaum's Outline of Complex Variables	Murray R. Spiegel, Seymour Lipschutz, John J. Schiller	McGraw-Hill
2	Higher Engineering Mathematics	B S Grewal	Khanna Publishers, New Delhi
3	Discrete Mathematics and Its Applications	Kenneth H. Rosen	Eighth Edition, McGraw-Hill Education
4	Schaum's outlines of Statistics, 6th Edition	Dr. Murray R. Spiegel & Larry J. Stephens	McGraw-Hill
5	A Textbook of Graph Theory 2nd Ed.	R. Balakrishnan, K. Ranganathan	Springer
6	A First Course at Graph Theory	John Clark and Derek Allan Holton	World Scientific

WEB RESOURCES		
Sl. No.	URL	Modules Covered
	NIL	

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50		60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

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